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DEPARTMENT OF COMPUTER ENGINEERING

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ECHELON: AI-Powered Predictive Maintenance for Industrial Equipment using XGBoost

SYNOPSIS

INTRODUCTION

Predictive maintenance is a data-driven approach used to identify the likelihood of equipment failure before an unexpected breakdown occurs. In conventional maintenance practices, equipment is typically serviced at arbitrary fixed calendar intervals (preventative) or only after catastrophic breakdown is observed (corrective). Such legacy approaches result in unnecessary maintenance costs, premature component disposal, or catastrophic operational downtime.

Machine learning methods support modern Industry 4.0 predictive maintenance by modeling the complex non-linear relationships between continuous operational parameters (temperatures, speeds, mechanical loads, wear) and observed machine degradation events.

The proposed case mini project, ECHELON, implements an end-to-end predictive-maintenance intelligence prototype for industrial equipment. Using the AI4I 2020 Predictive Maintenance Dataset, the project develops a calibrated Extreme Gradient Boosting (XGBoost) classifier to estimate failure probability from multidimensional sensor telemetry.

OBJECTIVES

- To study the application of machine learning to predictive maintenance of industrial equipment.
- To preprocess and analyze the AI4I 2020 Predictive Maintenance Dataset and identify relevant operating features for machine-failure prediction.
- To develop an XGBoost-based binary classification model for predicting machine failure from industrial operating parameters.
- To generate failure probability, machine health score and risk level from the trained model output.
- To develop an interactive dashboard that supports simulated sensor monitoring and manual machine-condition analysis.
- To implement a What-If Simulator for examining how changes in operating conditions affect predicted failure risk.
- To evaluate the model using accuracy, precision, recall, F1-score, ROC-AUC and confusion matrix.

MODULES

Module 1: Dataset Preparation and Analysis. Load and inspect the AI4I dataset, check data quality, analyze class imbalance, encode the categorical product-type feature, select relevant variables and prepare the data for training and testing.

Module 2: XGBoost Failure Prediction. Train the XGBoost classifier for binary machine-failure prediction, account for the minority failure class during training and store the trained model and preprocessing artifacts for inference.

Module 3: Machine Health and Scenario Analysis. Convert model probabilities into machine health score and risk level, provide rule-based maintenance advisories and implement the What-If Simulator for scenario comparison.

Module 4: ECHELON Dashboard and Evaluation. Develop the Streamlit dashboard for simulated sensor monitoring, manual analysis and model insights, and present the model performance using appropriate evaluation metrics and visualizations.

MATERIAL AND METHODS

1. Software Tools, Libraries, and Technology Stack

- **Language & Runtime:** Python 3.13
- **Machine Learning Engine:** XGBoost: Extreme Gradient Boosting Classifier.
- **Data Processing & Scientific Computing:** pandas, numpy
- **Machine Learning Utilities:** scikit-learn for stratified data splitting, LabelEncoder, StandardScaler, and evaluation metrics.
- **Visualization Layer:** plotly for interactive web charts; matplotlib (v3.10.8) and seaborn (v0.13.2) for static evaluation generation.
- **Application Framework:** streamlit with custom CSS styling

2. Dataset Description: The AI4I 2020 Predictive Maintenance Dataset comprises 10,000 records. Features include Type (Low/Medium/High), Air temperature (K), Process temperature (K), Rotational speed (rpm), Torque (Nm), and Tool wear (min). The target variable is Machine failure (Binary, 3.39% minority class).

3. Empirical Validation Results:

Evaluation Metric	Empirical Test Result	Operational Industrial Interpretation
Accuracy	98.15% (0.9815)	High overall correct classification rate across test partition.
Precision	70.67% (0.7067)	Low false alarm overhead; 7 out of 10 failure warnings are true defects.
Recall (Sensitivity)	77.94% (0.7794)	Successfully catches ~78% of all true machine failure occurrences.
F1-Score	74.13% (0.7413)	Strong harmonic balance between precision and sensitivity.
ROC-AUC	96.72% (0.9672)	Near-optimal discriminative separation across class distributions.

DURATION OF STUDY

Week 1: Data Preparation — Dataset ingestion, cleaning, feature engineering, and stratified train/test splitting.

Week 2: Model Training — Building the XGBoost classifier, addressing class imbalance, and hyperparameter tuning.

Week 3: Evaluation & Logic — Assessing model metrics (ROC-AUC, F1, Recall), formulating machine health scores, and defining rule-based advisories.

Week 4: Dashboard & Deployment — Developing the interactive Streamlit application, end-to-end testing, and finalizing documentation.

REFERENCES

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CONCLUSION

ECHELON demonstrates an applied predictive maintenance architecture that resolves the industrial class imbalance challenge through cost-sensitive gradient boosting. Achieving a 96.72% ROC-AUC and 77.94% recall, the model successfully detects failures with minimal false alarms. Beyond raw predictions, ECHELON delivers operational value through standardized Health Scores, deterministic advisory rules, and a 4-tier interactive dashboard, empowering supervisors to model risk impacts and mitigate unplanned downtime. Future enhancements may include integration with real-time industrial sensors or PLC systems, validation on real machine data, time-series analysis, richer failure-mode diagnosis, additional explainability methods and deployment in an actual industrial environment.